

Index, Retrieve, Generate

DATA 443/643 · Week 14, note 14.1

Week 14

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Ideas from Stanford CS224N 2026 L10 (RAG at inference). Original slides; that course is not copied.

Lecture plan

- 1 Non-parametric memory at request time
- 2 Cosine rank on three toy docs
- 3 $k=1$ vs $k=2$: near-miss d3
- 4 Not Week 10 RAFT

Key goal: recall@k and an attribution check, plus a no-RAG baseline.

Conditioning, not a new net

- Index chunks. Retrieve top-k. Stuff, then generate.
- CS224N L10: facts live in the index, not only in weights.
- Week 10.3 RAFT trains on those docs. This week is inference.

RAG: non-parametric memory at inference (and, in RAFT, at train time)



Bag-of-words cosine

$$\cos(q, d) = \frac{q^\top d}{\|q\| \|d\|}$$

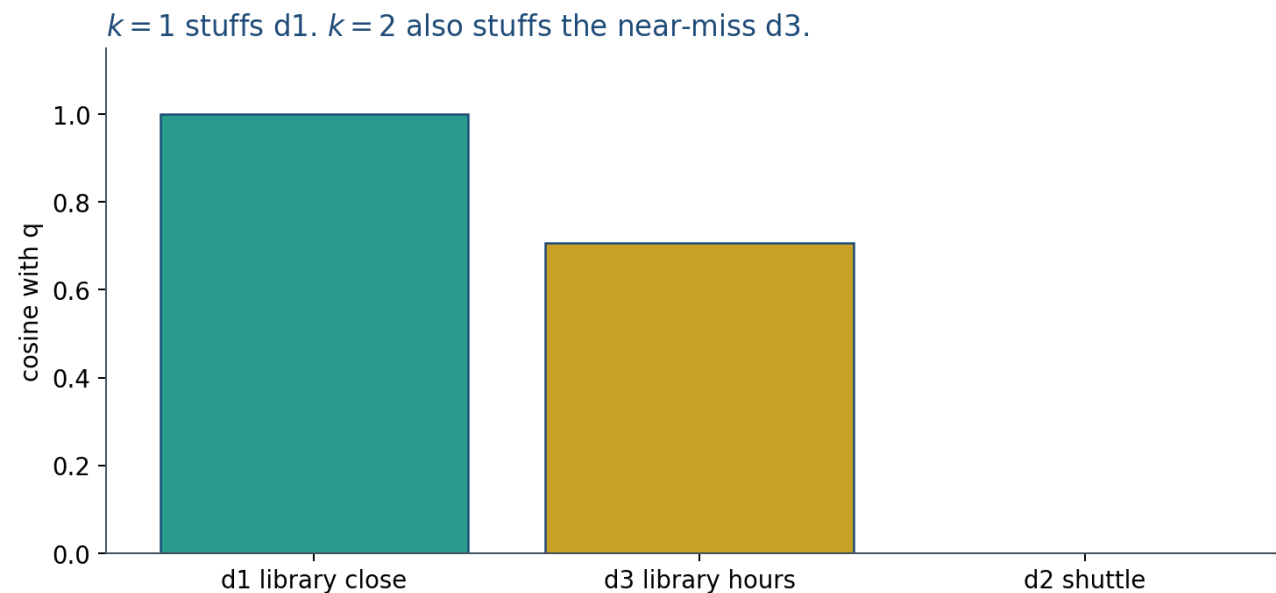
$q = d1 = (1, 1, 0)$: cosine
1 with itself.

$d3 = (1, 0, 0)$:
 $1 / \sqrt{2} \approx 0.707$. $d2$: 0.

Lab 14 prints this ranking. No GPU.

k is a method knob

- $k=1$ stuffs d1 (the close-time chunk).
- $k=2$ also stuffs d3 (hours). Recall can rise; the near-miss can mislead.
- Lost in the middle: gold in a long context is often ignored.



Two RAG weeks

10.3 RAFT

A training recipe: learn to use retrieved docs. Weights change.

14.1 this hour

The inference stack: index, retrieve, generate, cite.
Measure recall@k.

What to take from this lecture

Two metrics

Retrieval recall@k, then answer + attribution.

Cite ids

Vectors without raw text cannot be cited.

Lab 14

Five AU-toy snippets. Cosine, then a citation.